

Commitment-pool route graphs for finance and mutual aid

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Abstract

This paper develops a bounded operationalization framework for selected financial and mutual-aid institutions. It models Commitment Pooling Protocol (CPP) as a four-function operational structure: curation of admissible claims, valuation of those claims, limitation of flows, and exchange through fee and settlement logic.

The paper asks when an institutional mechanism can be realized through stable stewarded implementation interfaces without claiming legal, cultural, welfare, or historical equivalence. It formalizes a source model of finite states, actions, guarded transitions, priority or loss ordering, and settlement; a target model of commitment pools and routes; a realization mapping; and a bounded observational-equivalence criterion.

The main result is a constructive direct-realization theorem under finite-state, guard-specification, governance, routing, primitive-completeness, and settlement-compilability assumptions, with a bounded repair-extended proposition for modeled endogenous repair. Worked examples include deposits, repo-style secured lending, central counterparty clearing, rotating labor, rotating savings, grain banks, and reserve-support repair. A court-supervised restructuring case marks a direct-realization boundary.

Keywords: payment systems, clearing and settlement, mutual credit, formal modeling, community finance

1 Introduction

Financial institutions are often treated as distinct species: deposits, loans, repos, insurers, clearing houses, and funds. Mutual-aid institutions are similarly treated as a separate family: rotating savings associations, labor-sharing groups, grain banks, and burial societies. This paper asks a narrow technical question: can selected institutions from both families be represented in one operational basis without erasing legal and cultural differences?

The question is not whether all institutions are “the same.” It is whether their core operational mechanics can be encoded in a shared model with explicit state, transition guards, priority rules, and settlement operators.

The term *commitment pool* is used here in a specific operational sense. At the protocol level, CPP has four core functions: curation, valuation, limitation, and exchange. At the implementation level, these functions are exposed through six interfaces: Registry, Value Index, Limiter, Fee Policy, Vault/Settlement, and Governance. Registry implements curation; Value Index implements valuation; Limiter implements limitation; Fee Policy and Vault/Settlement implement exchange. In the formal model below, Governance appears as an implementation interface because authorized parameter changes must be represented. Conceptually, however, governance belongs to the stewardship environment together with authentication, memory, evidence, and dispute systems.

A pool is therefore a governed ledger or contract system that can list admissible claims, assign values, impose limits and fees, hold commitments or reserves, and execute settlement.

A pool need not be a large collective institution. At the smallest scale, an individual, household, or enterprise may operate a personal or organizational pool, provided that it exposes the relevant interfaces for admissibility, valuation, limits, exchange, settlement, and governance. A single promise by an individual is therefore a commitment; it becomes pool-like only when embedded in an explicit rule-governed interface for holding, accepting, routing, or settling commitments.

A *route graph* links pools when claims or settlement paths can move across more than one pool subject to each pool’s local rules. A *steward* is any actor, committee, multisignature group, governance process, or rule system authorized to update pool parameters such as admissibility, valuation, limits, fees, pause states, remedies, or routing permissions.

This perspective is related to layered-claims and balance-sheet approaches in monetary economics ([Ingham 2004](#); [Minsky 1986](#); [Mehrling](#)

2011), but adds explicit operational interfaces for a four-function commitment architecture. The framework is also informed by field-oriented and software-oriented work on community currencies, mutual credit, and commitment pooling (Ruddick 2023; Ruddick 2026; Grassroots Economics 2026a; Grassroots Economics 2026b; Grassroots Economics 2026c; Grassroots Economics 2026d; Cosmo-Local Credit 2026). Those implementations motivate the interface decomposition, but they are not evidence for the theorem. The formal claim below stands or fails on the stated assumptions, mappings, and observational-equivalence criterion.

Why commitment pools rather than a generic ledger or balance-sheet graph? The differentiator is not data storage; it is operationalization through policy decomposition around four questions: what may enter, how it is valued, how much may flow, and how exchange settles. This separation makes it possible to ask whether a specified institution’s states, actions, guards, priorities, settlement effects, and repair paths can be realized through fixed interface types without adding institution-specific primitives.

The contribution is therefore a bounded operationalization test, not a catalog of institutions that resemble commitment pools and not a claim that a ledger can encode arbitrary social arrangements. Institutions that appear legally or culturally distinct can be compared by asking which interface margins differ, which transitions fail, which repair path is invoked, and which observables are preserved. This creates a common representation for studying institutional interoperability, bounded liquidity access, settlement failure, and structured repair without asserting legal, cultural, welfare, or ontological equivalence.

For economics, the value of the framework is comparative and diagnostic. It does not replace models of incentives, welfare, or equilibrium. Rather, it provides a common operational substrate on which such questions can be posed more precisely. Once an institution is expressed through the pool-route interfaces, one can ask which margin changes across designs or over time: admissibility, valuation, limits, fees, custody, routing, governance, settlement, or repair. This makes institutional comparison less dependent on labels and more dependent on explicit mechanism structure.

Contributions. This paper contributes four items.

1. A formalization of institutional mechanisms as finite guarded transition systems over claims, reserves, collateral, priorities, settlement operators, and governance states.

2. A commitment-pool route target model that distinguishes four CPP functions (curation, valuation, limitation, and exchange) from their implementation interfaces (Registry, Value Index, Limiter, Fee Policy, Vault/Settlement, and Governance), while allowing stewarded parameter updates over time.
3. A constructive direct-realization theorem with explicit lemmas and preservation conditions, plus a bounded repair-extended proposition for endogenous repair.
4. A reusable operational realization test for deciding whether a proposed mechanism is directly realizable, repair-extended, externally anchored, or outside the framework’s present scope.

Roadmap. Section 2 positions related work. Section 3 defines source and target models, mapping, equivalence, repair classes, and failure boundaries. Section 4 proves direct realizability. Sections 5 and 6 provide worked constructions. Section 7 discusses limits, repair modes, and a negative case for direct realization. Section 8 connects theorem conditions to software verification obligations. Section 9 applies the operational realization test. Section 10 concludes. Appendix A gives a compact reusable test procedure, and Appendix B gives a fully enumerated memory-based ROLA witness.

2 Related Work and Positioning

2.1 Money and layered claims

Monetary hierarchy and layered-claim views emphasize that modern money is structured by issuer liabilities, settlement hierarchy, and convertibility constraints (Ingham 2004; Minsky 1986; Mehrling 2011). This literature explains why deposits, reserves, and collateralized liabilities are relational claims, but usually does not provide a single implementation schema spanning formal finance and mutual-aid institutions.

2.2 Institutional and commons governance

Commons research shows that governance quality, monitoring, and sanctioning determine performance of shared resource systems (Ostrom 1990). Anthropological and historical treatments of reciprocity and exchange (Mauss

2002; Polanyi 1944; Graeber 2011) clarify that social legitimacy and obligation cannot be reduced to prices alone. The framework treats these as governance and state constraints, not as exogenous commentary.

2.3 Market infrastructure and settlement

Market-infrastructure and funding-liquidity literature, including work on central counterparties (CCPs), identifies margining, netting, and settlement-waterfall design as core drivers of systemic outcomes (Duffie and Zhu 2011; Brunnermeier and Pedersen 2009; Committee on Payment and Settlement Systems and Technical Committee of IOSCO 2012). The present model contributes a uniform operational mapping language across these mechanisms. Related work on obligation-clearing and mutual credit also shows that liquidity-saving and multilateral clearing can be materially improved when obligation networks are combined with a complementary mutual-credit layer (Fleischman, Dini, and Littera 2020). At the level of formal method, the paper is also adjacent to state-transition and specification traditions that emphasize explicit system state, guarded actions, and observational reasoning (Lamport 2002; Clarke, Grumberg, and Peled 1999). The paper is also adjacent to Promise Theory, which models coordination among autonomous agents through promises rather than through externally imposed commands or obligations (Burgess and Fagernes 2006; Bergstra and Burgess 2008; Bergstra and Burgess 2014). In the present paper, a commitment is used in this promise-theoretic sense: an issuer makes a declared promise or attestation whose operational consequences depend on how pools curate, value, limit, hold, route, and govern it. The paper does not rely on Promise Theory’s full formal apparatus, but uses its agent-centered interpretation of promises to clarify that commitments are not assumed to be legally coercive obligations.

2.4 Contracts, mechanism design, and implementation

The paper is also related to contract theory and mechanism design, but it asks a different question. Contract theory studies enforceability, incompleteness, allocation of control rights, and optimal contracting under informational and enforcement constraints. Mechanism design studies whether desired social-choice rules can be implemented under strategic behavior. The present paper does not solve an optimal-contract, incentive-compatibility, or equilibrium-selection problem. It asks whether the operational state transitions of selected institutional mechanisms can be realized through

commitment-pool route interface types while preserving bounded external observables.

Incentive compatibility, welfare comparison, bargaining power, and equilibrium selection are therefore outside the theorem unless they are encoded directly into the source state machine, guard predicates, strategy space, and observable criteria. This distinction is important: a mechanism may be operationally realizable without being incentive-compatible, fair, efficient, legally enforceable, or normatively desirable.

2.5 Complementary currencies and mutual-aid institutions

Community-currency studies document practical trade-offs in liquidity, inclusion, and governance (Seyfang and Longhurst 2013; Lietaer et al. 2012). Classic and modern work on rotating savings and credit associations (ROSCAs) and related mutual-aid forms examines their comparative structure, economic role, and household allocation functions (Ardener 1964; Besley, Coate, and Loury 1993; Anderson and Baland 2002). Related labor-sharing institutions also deserve explicit notice. Rotating labor associations (ROLAs) and closely related reciprocal labor groups have been documented as structured mutual-help arrangements in which participants coordinate labor contributions across households or plots according to pre-established reciprocity or rotation rules (Wang 2019; Messerschmidt 1981). Prior work typically covers subsets (currency design, governance, or exchange rules) rather than a unified transition-system representation with explicit settlement and priority operators. Within the emerging Commitment Pooling literature itself, field-oriented and stewardship-oriented accounts already describe the protocol as both a practical coordination system and a commons-based infrastructure for reciprocity and mutual support (Ruddick 2023; Ruddick 2026). The present paper differs by asking a narrower formal question about realization, equivalence, and bounded theorem-level representability.

Interpretive scope note. Where kinship, ritual obligation, honor, gendered labor roles, or informal sanction matter causally, they are represented here only insofar as they alter guards, permissions, priorities, or remedies at the chosen modeling level. A successful realization therefore preserves operational sequence at that level of abstraction; it does not preserve full social meaning, legitimacy, or lived phenomenology. The choice of abstraction level is therefore substantive rather than neutral. If a kinship rule, ritual duty, gendered labor norm, elder decision, or informal sanction changes feasible

actions, priorities, settlement effects, or expected remedies, then omitting it from the modeled state, transition, priority, or settlement description is a modeling loss rather than a successful realization.

3 Formal Framework

3.1 Source model for institutional mechanisms

The source-model choice is deliberately state-machine based, in the spirit of formal specification frameworks that treat system behavior as guarded transitions over explicit state (Lamport 2002; Clarke, Grumberg, and Peled 1999).

Modeling note. The source model is stated in ordinary state-machine language: state, action, guarded transition, priority or loss ordering, and settlement operator. It does not use pool-specific interface terms such as registry, valuation policy, limiter, fee policy, custody, or governance as primitive terms. This separation is important because the realization result should not be obtained by defining the source institution in the same language as the target pool graph. Other institutional descriptions are in scope only insofar as they can be represented at the chosen granularity through finite states, explicit actions, guarded transitions, priority rules, and settlement effects.

The state representation need not be a written or digital ledger. In oral or memory-based institutions, recognized group memory, witness testimony, and steward attestation may constitute relevant state and evidence variables, provided that membership, contribution, obligation, dispute, and remedy effects are observable at the chosen level of abstraction.

Definition 1 (Institutional state machine). *An institution is a tuple*

$$\mathcal{I} = (S, A, T, \Pi, \Sigma),$$

where:

- *S is a finite state space of claim, reserve, collateral, and governance states.*
- *A is a finite action set available to participants and stewards.*
- *$T \subseteq S \times A \times \Gamma \times S$ is a guarded transition relation, with guard set Γ .*

- Π is a priority ordering over claim classes in stress/default states.
- Σ is a settlement operator that maps executable obligations and inventories to realized settlement outputs.

Settlement-layer note. The operator Σ is institution-relative. For a retail wallet, Σ may denote redemption into a designated settlement asset; for a CCP, novation plus default-waterfall execution; for a grain bank, inventory release subject to emergency-floor rules. The theorem therefore preserves settlement outputs relative to the source institution’s own settlement layer and legal design, not a universal notion of finality across all institutions. For a past-oriented certification, Σ may denote pool acceptance, exchange, or settlement into other commitments rather than redemption against the original issuer.

Definition 2 (State trajectory). *For horizon $H \in \mathbb{N}$ and participant strategy profile σ , a trajectory is*

$$\tau_H(\sigma) = (s_0, a_0, s_1, \dots, a_{H-1}, s_H),$$

with $(s_t, a_t, \gamma_t, s_{t+1}) \in T$ for admissible guards γ_t .

3.2 Target model (commitment-pool route graph)

Definition 3 (Commitment). *A commitment is a pool-admissible promise or attestation:*

$$C = (i, j, K, b, q, \chi, t, e, r, m),$$

where i is the autonomous issuer, promiser, or attestor, j is the promisee, recipient, holder, awardee, beneficiary, or certified subject, depending on the temporal orientation of the commitment, K is the scope or publication set of agents for whom the commitment is made visible, b is the promise body, q is quantity or contingent formula, $\chi \in \{\text{future}, \text{past}\}$ is temporal orientation, t specifies time conditions, event time, or validity window, e specifies evidence requirements, r is the remedy, fallback, revocation, or correction rule, and m is metadata.

Temporal-orientation note. If $\chi = \text{future}$, the commitment is promissory: it concerns a future behavior, service, redemption, delivery, or outcome. If $\chi = \text{past}$, the commitment is a certification: an issuer or attestor declares that a past event, state, contribution, delivery, or performance occurred. A certification is not treated as an objective fact by assumption; it

is a promise-like attestation whose admissibility depends on issuer standing, recipient or holder status, evidence, scope, valuation, limits, and governance rules.

A past-oriented certification need not be redeemable against its issuer in the way a voucher or future-oriented redemption promise may be. Nevertheless, it may still be tokenized, held, transferred, pooled, and accepted by a pool in exchange for other commitments or resources held by that pool, provided the pool’s curation, valuation, limitation, evidence, and settlement rules permit such acceptance. In this sense, both future-oriented promises and past-oriented certifications can function as pool-admissible commitments, but their settlement semantics may differ.

Definition 4 (Commitment pool). *A pool implementation is*

$$P = (R, V, L, F, H, G),$$

where R is the Registry, V is the Value Index, L is the Limiter, F is Fee Policy, H is Vault/Settlement custody state, and G is Governance permissions.

The tuple implements the four CPP functions: R implements curation, V valuation, L limitation, and (F, H) exchange. Governance G authorizes parameter changes and is treated as part of the stewardship environment rather than as a fifth protocol function. Registry entries, values, limits, fees, custody states, routing permissions, pause states, and governance permissions may change through authorized steward transitions during ordinary operation or repair.

Fee policy F may include local swap fees, route-hop fees, support fees, clearing fees, or interchange-like fee splits across multiple pools. These fees are evaluated as part of exchange feasibility and settlement output.

Definition 5 (Route graph). *A route graph is $G_P = (\mathcal{P}, E)$, where $\mathcal{P}$ is a set of pools and directed edges $e = (P_u, P_v)$ are admissible only if overlap and guard constraints are satisfied.*

At network scale, discovery registries and routing services are auxiliary coordination functions rather than separate claim primitives. They publish machine-readable pool metadata needed for interoperation, including accepted commitment classes, valuation rules, limit policies, fee schedules, pause states, and, where disclosed, recent exchange and remedy logs. When these interfaces are transparently broadcast on a shared ledger or comparable publication layer, admissible routes become discoverable and agents can

also evaluate pool standing, support capacity, and clearing opportunities across larger pool networks. These functions are not only conceptual: existing open-source stacks already separate pool logic, registries, valuation quoters, limiters, fee controls, and indexing or discovery services in ways broadly consistent with the present target-model decomposition (Grassroots Economics 2026a; Grassroots Economics 2026b; Grassroots Economics 2026c; Cosmo-Local Credit 2026).

3.3 Mapping and equivalence

Definition 6 (Realization mapping). *Let S_P denote the global state space of a target commitment-pool route graph. A mapping*

$$\Phi : I \mapsto G_P$$

realizes institution I if there exist:

1. *a state embedding $\alpha_\Phi : S \rightarrow S_P$ from source states into target route-graph states;*
2. *an observation projection $\omega_\Phi : S_P \rightarrow O$ into externally relevant observables;*
3. *a primitive translation schema ψ such that each enabled source transition in T , after factorization into primitives from U , is implemented by one local pool transition or a finite admissible route; and*
4. *preservation of source settlement and priority observables, so that target execution yields the same projected settlement, residual-obligation, and priority/loss outcomes as Σ and Π at macro-step boundaries.*

The embedding α_Φ is not required to preserve internal implementation structure. It is required only to preserve the observable institutional state selected by the model.

Definition 7 (Primitive translation schema). *Let*

$$U = \{\text{issue, transfer, collateralize, guarantee-invoke, settle, default/remedy}\}.$$

A translation schema ψ assigns each $u \in U$ a target implementation template

$$\psi(u) = (\Delta R, \Delta V, \Delta L, \Delta F, \Delta H, \Delta G, \rho),$$

where ρ is either empty or a finite admissible route.

Table 1: Primitive translation templates.

Primitive	Target interfaces affected	Required guards and outputs
issue	R, H, G	Issuer or attester permission; registry admissibility; creation or publication of claim or certification; receipt emitted.
transfer	R, V, L, F, H	Sender holding sufficiency; recipient or pool admissibility; value and fee calculation; limit check; custody update; receipt emitted.
collateralize	R, V, L, H, G	Collateral class listed; valuation rule valid; pledge permission; exposure or haircut constraint; pledged custody state updated.
guarantee- invoke	R, V, L, H, G	Guarantee listed; trigger condition satisfied; guarantor or reserve capacity available; payout or support obligation recorded.
settle	R, V, L, F, H, G	Settlement guard satisfied; inventory or reserve available; fees applied; claim extinguished, fulfilled, deferred, or left outstanding according to Σ ; receipt emitted.
default/remedy	R, V, L, F, H, G	Default or breach trigger satisfied; priority rule Π applied; remedy, pause, liquidation, relimit, support, or terminal state recorded.

Four-function interface restriction. A realization is admissible only if every target transition updates the fixed implementation interfaces (R, V, L, F, H, G) through the primitive templates in ψ . Equivalently, target transitions must operate by curation, valuation, limitation, or exchange as implemented by those interfaces. No institution-specific primitive may be added after observing the source mechanism. Institution-specific variation may enter only through parameters, guard predicates, valuation rules, limits, fee rules, custody states, route predicates, governance permissions, evidence conditions, and settlement rules already expressible through those interfaces.

This restriction prevents the result from becoming an unrestricted encoding theorem. Interface parameters may change over time, but the operational language is not redesigned for each case.

In this primitive basis, *issue* includes both the issuance of future-oriented commitments and the publication or award of past-oriented certifications. Pool acceptance of a certification in exchange for other commitments is represented through the existing *transfer* and *settle* primitives, subject to the pool’s registry, valuation, limitation, custody, and governance rules.

Let $\mathcal{S}_H$ denote the set of admissible participant and steward strategy profiles over horizon H .

Definition 8 (Stutter-insensitive observational equivalence over horizon H). *Given strategy space $\mathcal{S}_H$, two implementations X and Y are equivalent, denoted*

$$X \equiv_H Y,$$

if for every $\sigma \in \mathcal{S}_H$, the projected traces over externally observable variables are equal after quotienting out internal stutter steps and route-internal micro-transitions:

$$\text{Obs}^*(\tau_H^X(\sigma)) = \text{Obs}^*(\tau_H^Y(\sigma)).$$

Here Obs^ records externally visible settlements, residual obligations, and realized priority/loss outcomes at macro-step boundaries. In a realized target system, Obs^* is induced by the projection ω_Φ from Definition 6 after quotienting out internal stutter steps and route-internal micro-transitions.*

External observables exclude internal governance deliberation unless deliberation changes executable permissions, and exclude cultural meaning unless it changes observable actions, guards, or settlement outputs. A source strategy profile σ is compared to its target realization through the natural lifting induced by ψ , where each source macro-action is implemented by its

corresponding target local transition or finite route execution and internal micro-steps are hidden by Obs^* .

Why this equivalence notion? The intended claim is preservation of institutionally relevant outcomes at macro-step boundaries, not identity of internal microstructure. Inter-pool routing, internal rebalancing, and stewarded micro-steps may differ across realizations while leaving externally visible settlements, residual obligations, and realized priority/loss outcomes unchanged. For that reason the paper uses bounded stutter-insensitive observational equivalence rather than strong bisimulation. This choice weakens the equivalence notion and should be read accordingly: the theorem does not preserve internal timing, deliberation paths, computational complexity, welfare, or strategic incentives except where these are already encoded in the observable state.

Definition 9 (Primitive completeness of the source model). *A source institution is primitive-complete with respect to U if every enabled transition in T can be factorized into a finite sequence of parameterized primitives drawn from U .*

Non-vacuity note. The direct-realization theorem in Section 4 is not an unrestricted graph-encoding result. The target language is specified ex ante by the primitive basis U and the pool interface types (R, V, L, F, H, G) ; the proof therefore does not permit institution-specific target primitives to be introduced ad hoc for each case. Interface parameters may change through authorized state transitions, but the interface language itself is not redesigned for each case. The substantive burden is that enabled source transitions must factorize into this fixed basis and that settlement and priority observables must be preserved. Institutions whose decisive mechanics require non-factorizable discretion, non-compilable settlement, or extra-procedural override fall outside direct realization.

Definition 10 (Repair transition class). *Let*

$$U^r = \{\text{reroute}, \text{defer}, \text{renegotiate}, \text{revalue}, \text{relimit}, \\ \text{recurate}, \text{guarantor-invoke}, \text{insurance-invoke}, \\ \text{support-request}, \text{adjudicate}, \text{override}, \text{re-enter}\}.$$

A repair transition is a state transition invoked when ordinary execution cannot proceed because of breach, dispute, blocked routing, reserve insufficiency,

valuation contestation, governance-triggered exception, or trust breakdown. A repair transition may update route feasibility, time conditions, valuation state, limit state, curation state, remedy state, guarantor or insurance obligations, reserve-support state, or governance permissions.

Definition 10a (Repair-schema completeness). A source institution is repair-schema complete with respect to $U \cup U^r$ if every enabled breach, dispute, exception, or remedy trajectory relevant to the modeled horizon can be factorized into a finite sequence of parameterized operations drawn from $U \cup U^r$, together with explicit triggers, state updates, and re-entry conditions.

Repair may therefore act on the same four functions used in ordinary execution: re-curation in R , re-valuation in V , relimiting in L , and exchange repair through fee or support reallocation in F and custody, reserve, guarantor, or insurance movements in H . Steward-authorized override, deferment, or re-entry is recorded in G .

Definition 11 (Repair-aware realization classes). *An institution is directly realizable if it admits a realization under the direct-realization assumptions stated in Section 4. It is repair-extended if its routine mechanics are directly realizable but some decisive breach, dispute, exception, or remedy trajectories require explicit repair transitions not yet covered by the direct-realization theorem in Section 4. It is externally anchored repair if decisive repair authority is exercised by courts, sovereigns, or political institutions external to the present pool-route formalization, even though their interventions may still have observable effects on claim state.*

The direct-realization theorem in Section 4 proves only direct realization. The repair-extended and externally anchored classes introduced here are used below to classify harder cases without treating them as non-institutional or permanently beyond representation.

The diagram separates construction, projection, and classification steps so that the direct theorem and the broader repair-aware interpretation are not confused. Figure 1 summarizes the source-to-target mapping pipeline used in the theorem and classification logic.

Toy equivalence example. Consider (a) a bank-like deposit ledger and (b) a fully reserved e-money pool. If both have identical admissible actions (deposit, transfer, redeem), identical guard conditions (inventory and policy checks), and identical settlement outputs over horizon H , they are equivalent under $\equiv_H$ even if their legal wrappers differ.

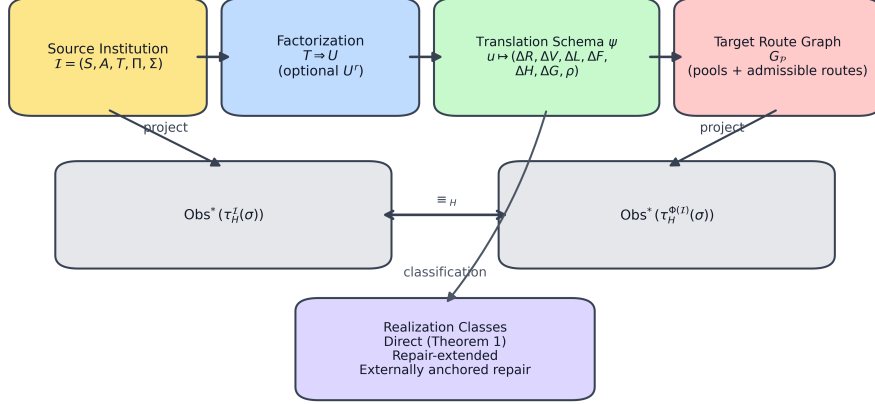


Figure 1: Source-to-target mapping pipeline. Direct realization follows the upper branch from source state machine to primitive factorization, primitive translation, and target route-graph execution. Repair branches support classification unless the relevant repair transitions are explicitly modeled.

3.4 Operational state variables

For reproducibility, we track:

1. Outstanding or registered commitments by class, including past-oriented certifications.
2. Custody balances and reserve balances.
3. Pledged collateral and margin utilization.
4. Limit utilization and breach flags.
5. Pending remedies/default status.
6. Governance parameter state and change log.
7. Published discovery metadata, route-fee schedules, pause states, and, where disclosed, fulfillment or breach history relevant to routing and pool standing.

3.5 Assumptions and failure boundaries

Assumption 1 (Finite-state adequacy). *The relevant institutional mechanisms can be represented at finite state granularity.*

This fails when obligations depend on unbounded discretionary narratives not reducible to finite guards.

Assumption 2 (Observable policy and repair interfaces). *Admissibility, valuation, limit, remedy, and repair policies can be encoded as explicit guard predicates or explicitly specified repair operations.*

This fails when key decisions require discretionary judgment that cannot be encoded as explicit predicates or as observable repair operations with specified triggers, state effects, and re-entry consequences.

Assumption 3 (Route composability). *Inter-pool transfers compose over finite admissible paths.*

This fails under hard legal partitions or forbidden interoperability boundaries.

Assumption 4 (Governance formalization). *Governance and emergency powers can be represented as state transition permissions.*

This fails when critical interventions are extra-procedural and not specifiable ex ante.

Assumption 5 (Settlement compilability). *For each in-scope claim class, the source settlement operator Σ can be compiled into target guard, custody, and payout logic without loss of relevant external observables.*

This fails when settlement depends on open-ended adjudication, non-codifiable evidentiary judgment, or extra-procedural intervention during execution.

Among these assumptions, Assumption 2 is usually the most restrictive. Many institutions fail direct representability not because claims and reserves cannot be modeled, but because acceptance, reprioritization, revaluation, limit adjustment, curation change, or remedy decisions depend on discretionary judgment that cannot yet be stabilized as explicit predicates or repair operations. Cases that violate direct Assumption 2 need not be dismissed as non-institutional. They may still be candidates for repair-extended or externally anchored repair representation if their repair processes have observable triggers, state effects, support paths, and re-entry consequences, even when those processes are not yet covered by the direct theorem.

Operational falsification criterion. A proposed in-scope mechanism falsifies the direct-realization claim if at least one enabled source transition cannot be factorized into the fixed primitive basis U , if a required state variable cannot be represented through (R, V, L, F, H, G) , if the settlement operator cannot be operationalized through custody, payout, route, and receipt logic, if priority or loss observables differ after projection, or if a decisive governance or repair step depends on discretion that cannot be represented as an explicit state transition over the modeled horizon.

Thus the framework is not validated by naming familiar institutions. It is tested by applying the realization procedure to a specified mechanism and checking whether the mechanism passes or fails the stated conditions.

4 Realization Theorem

Operationalization procedure. For an in-scope institution $\mathcal{I} = (S, A, T, \Pi, \Sigma)$, direct realization proceeds as follows.

1. Enumerate the relevant source state variables, actions, guards, priority rules, and settlement effects at the chosen level of abstraction.
2. Factor each enabled source transition in T into a finite sequence of primitives in U .
3. Instantiate the primitive translation schema $\psi(u)$ for each primitive $u \in U$.
4. Verify that each instantiated target transition updates only the stable interface types (R, V, L, F, H, G) and their authorized state variables.
5. Verify that route execution, where used, passes local registry, valuation, limit, fee, custody, inventory, and governance checks at every hop.
6. Compare the projected source and target trajectories over horizon H under Obs^* .
7. Classify any failed factorization, non-operationalizable settlement rule, unpreserved priority rule, or non-explicit discretionary repair step as a boundary case rather than as direct realization.

This procedure is constructive. It gives a method for building a realization when the conditions hold and a method for locating the failure point when they do not.

Lemma 1 (Primitive realization). *Every primitive source transition in U appearing in a factorization of T (issuance, transfer, collateralization, guarantee invocation, settlement, default/remedy) can be implemented by either a local pool transition or a finite guarded inter-pool route.*

Proof. For each primitive class $u \in U$, define $\psi(u)$ explicitly. Issuance updates registry admissibility and holdings state. Transfer updates holdings locally or across a feasible finite route. Collateralization updates pledged-holdings and limitation state. Guarantee invocation updates remedy state and priority constraints. Settlement updates executable obligations through custody state and Σ -consistent output rules. Default/remedy updates governance-authorized fallback transitions. Table 1 specifies the target implementation form for each primitive in U . Each template updates only the stable interface types (R, V, L, F, H, G) and requires explicit guards and receipt-producing state updates. Therefore, for every primitive source transition admitted by a valid factorization, there exists a corresponding local pool transition or finite guarded route. $\square$

Lemma 2 (Compositional closure). *If transitions u and v are realizable under Φ , then any finite composition $u; v; \dots$ is realizable by finite composition of corresponding pool/route transitions.*

Proof. Realizable transitions are represented as guarded relations over finite states. For any valid source factorization, the post-state of each primitive transition is the pre-state for the next primitive transition in the sequence. The corresponding target transitions are instantiated by ψ with guards evaluated against the target state induced by the source embedding and prior target updates. Therefore, whenever the next source primitive is enabled, the corresponding target primitive is enabled by the translated guard conditions. Guarded relation composition is closed under finite composition, and route concatenation is finite by Assumption 3. Therefore finite source sequences remain realizable. $\square$

Lemma 3 (Priority preservation). *If source priority ordering Π is encoded as explicit waterfall constraints in target guard and settlement logic, then realized loss-allocation order is preserved.*

Proof. Waterfall constraints enforce admissible payout order as hard transition guards. Any violating transition is infeasible. Therefore every feasible settlement path respects Π . $\square$

Theorem 1 (Constructive realization under assumptions). *Under Assumptions 1–5 and primitive completeness with respect to U , any institution $\mathcal{I} = (S, A, T, \Pi, \Sigma)$ whose mechanics are fully expressed by those components admits a finite commitment-pool route realization $\Phi(\mathcal{I})$ such that for any bounded horizon H , observables are preserved: $\Phi(\mathcal{I}) \equiv_H \mathcal{I}$.*

Proof. Induct on the length k of the primitive realization sequence obtained by factorizing enabled source transitions into elements of U .

Base case $k = 1$: realizable by Lemma 1.

Inductive step: assume realizability for primitive sequences of length k . For a sequence of length $k + 1$, compose one additional primitive transition. By Lemma 2, composition remains realizable. Any additional route-internal micro-steps introduced by composition are hidden by the Obs^* projection and therefore do not alter macro-step observables.

Priority/loss ordering is preserved by Lemma 3. Settlement outputs are preserved because the realization mapping requires Σ -consistent guard, custody, payout, and receipt logic for each in-scope claim class. Assumption 5 ensures that this compilation does not lose the relevant external observables. Therefore, after applying the observation projection in Definition 6 and quotienting out route-internal micro-steps, trajectory observables are preserved for all bounded horizons. $\square$

Scope note. Theorem 1 covers direct realization under explicitly formalized stewardship permissions. It does not yet prove realization for cases in which stewardship operates through open-ended adjudication, negotiated restructuring, contested re-valuation, limit reconstitution, emergency re-curation, insurance or guarantor support under contested conditions, network-level rerouting, or externally anchored override during execution, unless those repair operations are themselves formalized as observable state transitions. This is an existence-by-construction result at the chosen abstraction level. It does not establish uniqueness, efficiency, incentive compatibility, welfare superiority, or institutional desirability of the realizing route graph.

What the theorem proves. Theorem 1 is a conditional compilation result. It proves that once a source institution has been specified independently as a finite guarded state machine, and once its enabled transitions factor through the fixed primitive basis U , there exists a commitment-pool route realization that preserves the selected external observables over

bounded horizons. It does not prove that all financial or mutual-aid institutions satisfy these assumptions, that such realization is unique, or that the target realization preserves legal status, cultural meaning, welfare, incentive compatibility, or social legitimacy. The theorem therefore establishes bounded operational realizability, not institutional equivalence.

Proposition 1 (Bounded repair-extended realization for endogenous repair). *Under Assumptions 1–5, primitive completeness with respect to U , and repair-schema completeness with respect to $U \cup U^r$, any institution $\mathcal{I} = (S, A, T, \Pi, \Sigma)$ whose routine execution is directly realizable and whose modeled repair trajectories use only repair operations in U^r with triggers, permissions, state effects, and re-entry conditions encoded in the target governance state G admits a bounded-horizon repair-extended realization $\Phi^r(\mathcal{I})$ such that observables are preserved up to stutter-insensitive macro-step equivalence over the modeled horizon.*

Proof. Factor routine prefixes into primitives from U and repair segments into operations from U^r . By Theorem 1, routine segments are realizable. By repair-schema completeness, each modeled repair segment is represented by an explicit finite sequence of repair operations with observable triggers, state effects, and re-entry conditions. Since both routine and repair segments are finite guarded state transitions, and since repair-schema completeness requires explicit triggers, permissions, state effects, and re-entry conditions, the same induction used in Theorem 1 applies to the concatenated $U \cup U^r$ sequence. Observables are compared only at macro-step boundaries, so bounded composition preserves Obs^* over the modeled horizon. $\square$

5 Worked Constructions: Finance

This section gives explicit witnesses rather than classificatory labels. Sections 5.1–6.3 are direct cases under Theorem 1. Section 6.4 gives a bounded repair-extended witness aligned with Proposition 1. Externally anchored repair cases are classified in Sections 7 and 9, but are not yet proved here by end-to-end witness construction.

5.1 Deposit-transfer-redemption institution

Source machine. State includes customer balances, reserve balance, and pending redemptions. Actions are deposit, transfer, and redeem. Guards require nonnegative balances and reserve sufficiency for redemption.

Mapping.

- R : accepted deposit claim classes.
- V : fixed 1:1 index to settlement unit.
- L : transfer/redemption caps and anti-money-laundering (AML) policy flags.
- H : reserve custody and balance updates.
- G : emergency pause and parameter updates.

Invariants. Conservation and nonnegative balances. Settlement outcomes match source machine under $\equiv_H$.

Example trace. Source trajectory over horizon $H = 3$:

$$s_0 : \text{reserve} = 100, \text{bal}(A) = 0, \text{bal}(B) = 0$$

$$a_1 = \text{deposit}(A, 10) \Rightarrow s_1,$$

$$a_2 = \text{transfer}(A, B, 4) \Rightarrow s_2,$$

$$a_3 = \text{redeem}(B, 3) \Rightarrow s_3.$$

Target realization uses $\psi(\text{issue})$, $\psi(\text{transfer})$, and $\psi(\text{settle})$: reserve custody increases by 10 and issues a claim to A ; holdings move 4 from A to B ; redemption guard checks reserve sufficiency then settles 3 to B . Observable balances and settlement outputs match at each step under $\equiv_H$.

5.2 Secured lending / repo-style institution

Source machine. States track collateral inventory, haircut-adjusted exposure, margin calls, and default states. Actions: open position, mark-to-market, margin call, unwind.

Mapping. Collateral commitment classes enter R ; haircuts and marks in V ; exposure caps in L ; pledged collateral in H ; default and close-out waterfall in G plus Π encoding.

Failure path. If collateral value falls below thresholds and margin not posted, remedy transition triggers collateral liquidation according to encoded priority order.

Example trace.

s_0 : collateral value = 120,
 cash advanced = 100,
 haircut = 20,
 margin status = compliant.

Open repo position $\Rightarrow s_1$. Mark-to-market shock reduces collateral value to 105 $\Rightarrow s_2$. Margin call issued; if unmet, default/remedy transition liquidates collateral according to encoded close-out waterfall $\Rightarrow s_3$. Observable exposure, margin-breach status, and liquidation ordering match the source-machine trajectory. A margin breach can be written as

$$V_t(1 - h) < E_t,$$

where V_t is marked collateral value, h is the haircut, and E_t is current exposure. The margin-call transition either restores

$$V_t(1 - h) \geq E_t$$

through additional collateral or moves the state into default/remedy. Thus the trigger guard is explicit rather than left as an informal stress event.

5.3 CCP-style clearing witness

Source machine. States: member obligations, initial/variation margin, default fund, loss mutualization stage. Actions: novation, netting, margin updates, default waterfall.

Mapping. Novation and netting are represented by clearing-pool transitions; margin and default-fund balances in custody state; waterfall order encoded in priority guards. Preservation of waterfall order follows Lemma 3.

Example trace. Two members novate offsetting obligations into the clearing pool. Initial margin and variation margin are updated. One member defaults; the system applies margin, then default fund, then loss mutualization in declared order. Observable loss allocation matches Π by construction. For a defaulting member i , let M_i denote available margin, D_i the defaulter's default-fund contribution, D_{-i} the mutualized default fund of

non-defaulting members, and L the close-out loss. The waterfall applies losses in the ordered sequence

$$M_i \rightarrow D_i \rightarrow D_{-i},$$

subject to non-negativity and exhaustion constraints. A target transition is admissible only if no later layer is debited while an earlier layer has positive available balance. This gives a concrete target-side encoding of Π for the CCP witness. More generally, the same network representation can encode card-network-style payment rails in which authorization, routing, interchange-like fee allocation, and deferred settlement occur across multiple issuers, acceptance points, and settlement pools.

6 Worked Constructions: Mutual Aid

6.1 ROLA witness

Source machine. State: member contribution record or recognized group memory, rotation order or reciprocal claim status, task equivalence class, attendance status, arrears or make-up obligation, and completion status. Actions: contribute labor, allocate the current labor round to the designated beneficiary, record shortfall, and apply a predefined arrears or make-up remedy.

Mapping. Accepted labor-claim classes and task categories enter R ; task-equivalence or conversion rules enter V when heterogeneous work must be normalized; round order, attendance bounds, and maximum pull conditions enter L ; completed labor claims, make-up obligations, and any support labor entered from overlapping pools enter H ; stewarded remedy, temporary deferment, and re-entry rules enter G .

Example cycle. Four households agree to contribute one workday each in rotating order to prepare fields before planting. In round one, all four members contribute labor to household A 's plot. In round two, household C fails to appear for household B 's turn, creating a make-up obligation. Governance records the shortfall and applies the predefined remedy: either deferred labor contribution in the next round or a substitute service-equivalent contribution according to the task-normalization rule. Observable completed labor, residual obligations, and turn order match the source-machine trajectory.

This witness captures the operational core of a rotating labor association: scheduled contribution, rotating or reciprocal allocation of labor benefit, and predefined remedy for shortfall. It does not claim cultural or normative equivalence to all labor-sharing institutions; it shows only that the core labor-contribution, allocation, and remedy dynamics admit a direct operational realization when task classes, attendance rules, and remedies are explicit (Wang 2019; Arciniegas, Dumas, and Fahn 2025; Messerschmidt 1981). A completed labor contribution may also generate a past-oriented certification: for example, a steward or beneficiary may attest that a member contributed one workday. Such a certification need not be redeemable against the attestor, but it may be accepted by a pool in exchange for other commitments when the pool’s valuation, limit, evidence, and governance rules permit. Appendix B gives a fully enumerated memory-based ROLA witness in which there is no written ledger and elders serve as stewards over recognition, dispute, remedy, and re-entry.

6.2 ROSCA/VSLA witness

Source machine. For a rotating savings and credit association (ROSCA) or village savings and loan association (VSLA), state includes member contribution records, turn order, arrears, and payout status. Actions: contribute, allocate turn payout, record arrears remedy. The ROSCA component of this witness is consistent with comparative and economic literatures that treat rotating savings and credit associations as structured contribution-and-allocation systems rather than merely informal social groups (Ardener 1964; Besley, Coate, and Loury 1993; Anderson and Baland 2002).

Mapping. Turn-based limiter is L ; contribution/payout commitments in R and H ; sanction/remedy policies in G . Local social sanction norms remain external primitives.

Example cycle. Four members each contribute one unit per round with fixed ex ante turn order. After three successful contribution actions and one arrears event, the limiter blocks payout reordering and governance applies the predefined arrears remedy. Observable payout order and residual obligations match the source-machine trajectory. This witness abstracts from the full social richness of ROSCAs, but it preserves the core operational features emphasized in the literature: scheduled contribution, rotating allocation, and sanction or remedy against arrears (Ardener 1964; Besley, Coate, and

[Loury 1993](#)). This operational witness does not claim cultural or normative equivalence to any particular ROSCA or VSLA; it shows only that the core contribution, allocation, and remedy dynamics admit a direct operational realization.

6.3 Grain-bank witness

Source machine. State: inventory by season, household claims, emergency reserve floor. Actions: deposit grain, issue receipt claim, redeem receipt, emergency allocation.

Mapping. Inventory-backed claims in R , seasonal valuation in V , anti-depletion floor in L , storage custody in H , steward override for emergencies in G .

Example cycle. Households deposit 50 units of grain after harvest and receive receipt claims. Later, they redeem 20 units during a lean season. If inventory falls to the emergency floor, limiter rules block ordinary redemption and steward override authorizes emergency allocation only. Observable inventory and redemption outcomes match the source machine.

6.4 Repair-extended witness: cross-pool reserve shortfall

Source machine. State: household claims, local reserve balance, support eligibility, temporary redemption limits, and re-entry status. Actions: redeem, detect reserve shortfall, request support, receive support, relimit, re-enter ordinary redemption.

Mapping. Receipt claims remain in R ; ordinary and emergency valuation rules in V ; temporary redemption throttles in L ; support or reserve movements in H ; steward-authorized support request, relimit, and re-entry in G .

Example repair trace. A household presents a valid redemption claim, but the local pool fails reserve sufficiency at s_1 . The system enters repair mode and invokes support-request, followed by either reserve support from an overlapping pool or a temporary relimit transition. Once support arrives or the temporary limit state is resolved, the system invokes re-enter and completes redemption. Observable claim status, inventory effects, and

residual obligations match the source-machine repair trajectory at macro-step boundaries.

This witness is not a proof of universal repair-extended realization. It shows that explicitly modeled endogenous repair operations can be given the same kind of end-to-end operational treatment as direct cases.

7 Limits, Repair Modes, and Direct-Realization Boundary

7.1 External primitives

Tax authority, legal personality, bankruptcy stays, and judicial discretion remain external primitives. The framework does not claim to generate sovereign power. This is especially salient for sovereign guarantee facilities, where activation, repricing, and fiscal backstop management are often embedded in broader public-finance and contingent-liability frameworks rather than purely operational rule systems ([Razlog et al. 2020](#)).

7.2 Hard cases and repair modes

Representability weakens when quality is non-contractible, contingencies are semantically ambiguous, underwriting is purely discretionary, risks are highly correlated, or governance capture dominates formal rules. Some hard cases may be better understood not as non-representable institutions, but as institutions operating in repair mode rather than routine execution mode. In a network of pools, repair is not limited to adjudication or rerouting. It may occur through rerouting, deferred settlement, guarantor invocation, insurance payout, negotiated remedy, re-curation of accepted commitments, re-valuation of claims, limit tightening or relaxation, reserve support from overlapping pools, or governance-mediated override and re-entry. Routing is therefore not only a repair path. In ordinary execution it is the mechanism by which heterogeneous pools interoperate, share liquidity, and clear compatible obligations across a wider network. Under stress, the same routing layer can also function as a safeguard against misuse, enclosure, or localized capture by allowing participants to reroute around degraded pools, tighten acceptance conditions, or shift toward pools with stronger published standing and support capacity. In this sense, repair is not external to the protocol architecture: it acts through the same interfaces of curation, valuation, limitation, fee or support allocation, custody, and governance that structure ordinary execution. A broken commitment does not necessarily terminate

the institution; in many mutual-aid, rotating labor, and labor-sharing settings it moves the system from routine execution into repair mode, where sanctions, support, repricing, rescheduling, or pool reconfiguration restore an executable path. This is consistent with institutional accounts that treat monitoring, sanction, and rule adaptation as part of system persistence rather than as failures outside the institution (Ostrom 1990). This interpretation also aligns with prior Commitment Pooling work that treats rotational labor, reciprocity, mutual support, and stewardship as constitutive elements of the infrastructure rather than as informal residues outside it (Ruddick 2023; Ruddick 2026). Figure 2 summarizes this repair topology schematically by showing how a failure trigger can branch into multiple repair operations and then return, where possible, to ordinary execution.

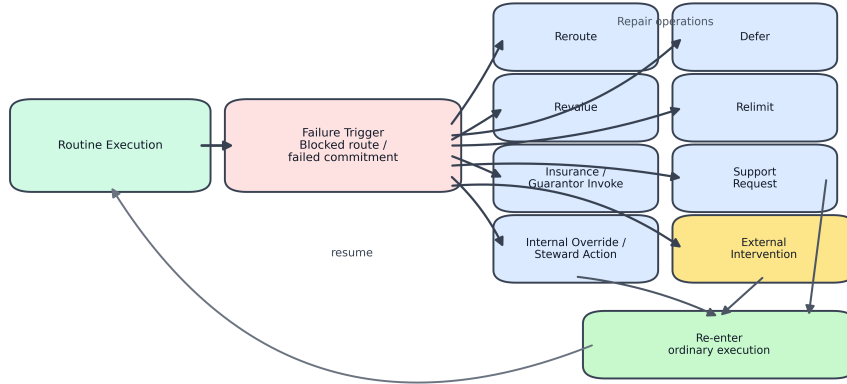


Figure 2: Repair topology in a network of pools. A failed routine transition may enter repair through rerouting, deferment, revaluation, re-curation, relimiting, guarantor or insurance support, stewarded override, external intervention, and possible re-entry into ordinary execution. The figure is diagnostic, not a proof sequence.

7.3 Negative case for direct realization

Consider a court-supervised restructuring process in which, during proceedings, the court may modify the stay, approve debtor-in-possession financing that primes existing claims, alter class formation, accept contested valua-

tion arguments, approve cramdown terms, or apply equitable subordination and public-interest reasoning without ex ante codified predicates. In such a process, admissibility, priority, and remedy are not fixed entirely by explicit guards prior to execution; they are partly produced by open-ended legal reasoning, litigation, bargaining, and discretionary intervention during the process itself. This characterization is consistent with standard treatments of bankruptcy as a process in which priority, control, financing, and restructuring outcomes may be altered through court-supervised legal procedure rather than fixed ex ante execution rules (Baird 2022). This directly violates Assumption 2 and may also violate Assumption 4 when decisive interventions are not representable as pre-specified state-transition permissions. For that subclass of institution, direct realization under Theorem 1 is not guaranteed. By contrast, narrower restructuring workflows with fixed voting, valuation, payout, support, and re-entry rules may still be directly or repair-extended representable. More generally, this does not imply that such cases are beyond institutional representation altogether; rather, they are better understood as candidates for externally anchored repair unless adjudicative, support, and repair transitions are themselves formalized within the model.

8 Software Verification Conditions

8.1 From theorem conditions to machine-checkable properties

Each lemma induces implementation obligations:

1. Primitive realization $\Rightarrow$ complete transition-handler coverage.
2. Compositional closure $\Rightarrow$ route composition cannot bypass limits.
3. Priority preservation $\Rightarrow$ executable waterfall proofs for all default paths.

These conditions should be read as proof obligations generated by the theorem, not as a claim that existing software artifacts already satisfy formal verification for all mapped institution classes. These are necessary but not sufficient verification conditions. Repair-aware implementations also require verifiable triggers, state updates, and audit trails for re-valuation, relimiting, support invocation, and override operations. A full implementation proof also requires settlement-operator conformance and evidence/remedy conformance for each mapped claim class.

Minimal invariant suite. Any implementation claiming conformance to a mapped institution should expose tests or proofs for the following invariants.

1. **Curation soundness.** Only admissible commitments may enter, move, or settle under the active registry, evidence, pause, and governance guards.
2. **Valuation soundness.** Quotes, haircuts, fees, and settlement amounts must follow the active value index and fee policy.
3. **Limitation soundness.** No transition or composed route may bypass balance, reserve, inventory, collateral, exposure, window, or global limit constraints.
4. **Exchange conservation.** Balances, commitments, reserves, collateral, inventories, and fee accounts must remain conserved and non-negative unless the source settlement operator explicitly permits signed balances, expiry, cancellation, burn, or retirement.
5. **Priority and settlement conformance.** Payout, extinguishment, residual-obligation, waterfall, and receipt effects must match Π and Σ after projection.
6. **Repair auditability.** Every repair transition must have an explicit trigger, permission source, state update, receipt, and re-entry or terminal condition.

These invariants are necessary verification obligations generated by the operationalization theorem. They are not sufficient for full economic, legal, security, or welfare validation.

8.2 Route selection formalism

Let the route graph be $G_R = (N, E)$, where N is the node set of pools, with edge feasibility predicate $\mathcal{F}(e, s)$ and cost $c(e, s)$. A path $p = (e_1, \dots, e_k)$ is feasible if there exists a state sequence $(s_1, \dots, s_{k+1})$ such that $\mathcal{F}(e_j, s_j) = 1$ for all $j \in \{1, \dots, k\}$, with s_{j+1} given by the state update induced by traversing edge e_j from state s_j . Optimization objective:

$$\min_p \sum_{j=1}^k c(e_j, s_j)$$

subject to feasibility and settlement-completion constraints.

Here $c(e, s)$ may include not only direct transfer cost, but also route-level fees, latency, inventory cost, support cost, or policy penalties. When pool interfaces and exchange logics are transparently published in machine-readable form, routing becomes discoverable rather than purely ad hoc, and the same formalism can support both point-to-point settlement and larger-scale credit clearing across federated networks of pools. This is consistent with prior work showing that obligation-clearing and mutual credit can jointly produce substantial liquidity-saving effects in small and medium-sized enterprise (SME) trade-credit networks (Fleischman, Dini, and Littera 2020). Failure occurs when no feasible path exists or no admissible path clears at acceptable cost. In the repair-aware reading of the framework, recovery policy is itself a repair operation and may select fallback assets, rerouting, deferment, support request, guarantor or insurance invocation, temporary revaluation, relimiting, or delayed settlement windows depending on state.

9 Operationalization Test and Illustrative Applications

This section applies the institution-independent operational realization test reported in Appendix A. The purpose is not to estimate the prevalence of directly realizable institutions. It is to show how the framework functions as a diagnostic procedure: given a proposed institutional mechanism, the test identifies whether it is directly realizable, repair-extended, externally anchored, or outside the model’s present scope.

The classification is mechanism-specific rather than label-specific. For example, a “loan,” “insurance pool,” “labor association,” or “clearing arrangement” may be directly realizable in one design and repair-extended or externally anchored in another, depending on whether states, guards, priorities, settlement effects, and repair authority are explicit.

Illustrative use. Table 2 reports a small set of applications. These examples are not part of the theorem and are not a statistical sample. They are included only to demonstrate how the same operational test can be applied across heterogeneous mechanisms.

Table 2: Illustrative applications of the operational realization test.

Specified mechanism	Classification	Reason for classification
Fully reserved retail deposit or wallet ledger with explicit redemption rule	Direct	Issuance, transfer, redemption, reserve custody, and settlement guards are explicit.
Repo with fixed haircut, margin-call trigger, and close-out rule	Direct	Collateral pledge, haircut, margin call, default trigger, and liquidation rule are codifiable.
CCP service with explicit margin and default-waterfall rules	Direct	Novation, margining, default fund, and loss waterfall can be represented through explicit priority and settlement rules.
Fixed-order ROSCA with explicit contribution, payout, and arrears rule	Direct	Contribution schedule, turn order, membership, and arrears remedy are explicit.
VSLA lending workflow with predefined default remedy	Direct or repair-extended	Routine lending may be directly realizable; renegotiation, social recourse, or discretionary forgiveness requires repair primitives if not fully specified.
Grain bank with seasonal receipts, reserve floor, and emergency allocation rule	Direct	Seasonal inventory, redemption windows, reserve floors, and emergency limits are operationalizable.
Memory-based rotating labor association with task-equivalence, elder recognition, and make-up labor rule	Direct or repair-extended	Turn order, labor-equivalence rules, and recognized contribution may be direct; contested memory, elder-mediated exceptions, absence, or make-up labor may require repair transitions.

Specified mechanism	Classification	Reason for classification
Private mutual insurance overlay with explicit trigger and payout rule	Repair-extended	Triggered payout can be modeled as repair; discretionary support or sovereign backstop is externally anchored.
Court-supervised bankruptcy with open-ended judicial discretion	Externally anchored	Stay scope, class formation, cramdown, and reprioritization may depend on open-ended judicial discretion.
Discretionary sovereign guarantee facility	Externally anchored	Activation, repricing, extension, and payout authority may depend on political or budgetary discretion outside pool permissions.

The examples show how the test locates the relevant boundary. A failed case is not necessarily outside institutional representation altogether. It may instead require explicit repair primitives, a narrower mechanism specification, or recognition that decisive authority lies outside the modeled pool-route system.

10 Conclusion

This paper advances a scoped bounded-operationalization claim for selected financial and mutual-aid institutions. It formalizes source and target models, a realization mapping Φ , bounded observational equivalence, repair-aware realization classes, a direct-realization theorem, and a bounded repair-extended proposition.

The main contribution is not that familiar institutions can be renamed as commitment pools. It is a test for when a specified mechanism can be operationalized through CPP’s four functions (curation, valuation, limitation, and exchange) and their fixed implementation interfaces without hidden primitives, unstated discretion, or loss of relevant observables. The framework is therefore a diagnostic and construction tool: it shows how selected mechanisms can be represented, where representation fails, which assumptions fail, and what repair or external authority must be specified before realization can be claimed.

The claim remains falsifiable. One robust in-scope mechanism that satisfies the stated scope conditions but cannot be realized through the fixed primitive basis and interface types is sufficient to reject or narrow the current theorem. Future work should convert more repair-extended cases into full witnesses and test whether this representation improves simulation, audit, routing, settlement, and institutional comparison in real systems.

A Operational Realization Test

This appendix gives a compact test for applying the framework to a proposed mechanism. The unit of analysis is not an institutional label such as “repo,” “ROSCA,” or “grain bank,” but a specified institutional mechanism

$$I = (S, A, T, \Pi, \Sigma).$$

A mechanism is directly realizable only if its relevant states, actions, guards, priorities, and settlement effects can be stated before introducing the target pool-route mapping.

Test step	Required specification
State	Claims, holders, reserves, collateral, inventories, limits, fees, permissions, disputes, defaults, evidence, and repair states relevant at the chosen abstraction level.
Actions	State-changing operations such as issue, transfer, redeem, pledge, margin, net, pause, revalue, invoke guarantee, reroute, settle, repair, or exit.
Guards	Conditions under which each action is allowed: membership, registry listing, balance sufficiency, limit availability, fee payment, inventory availability, collateral adequacy, governance approval, or evidence submission.
Priority	Any ordering over claims, losses, remedies, payouts, or settlement layers.
Settlement	What changes hands, what is extinguished, what remains outstanding, what receipt is emitted, and what happens when ordinary settlement fails.
Target mapping	Mapping into CPP functions and implementation interfaces (R, V, L, F, H, G) .
Primitive factorization	Verification that each enabled transition can be expressed through the fixed primitive basis U , without adding institution-specific primitives.
Route check	Verification that each route hop satisfies local registry, valuation, limit, fee, custody, inventory, pause-state, and governance constraints.
Observable equivalence	Equality of externally relevant settlements, residual obligations, and realized priority/loss outcomes over horizon H after projection by Obs^* .
Repair classification	If ordinary execution fails, specification of repair trigger, permission source, state update, receipt, and re-entry or terminal condition.

Table 3: Compact operational realization checklist.

A mechanism is *directly realizable* if all ordinary transitions pass the checklist. It is *repair-extended* if ordinary execution is directly realizable but breach, dispute, exception, or failure handling requires explicit repair primitives. It is *externally anchored* if decisive state changes depend on authority outside the modeled pool system, such as open-ended judicial, sovereign, regulatory, or political discretion. It is *outside scope* if the relevant mechanics, observables, or finite-state abstraction cannot be specified at the chosen level.

B Fully Enumerated Witness: Memory-Based ROLA

This appendix applies the operational realization test to a memory-based rotating labor association (ROLA) in which there is no physical or digital ledger. The witness is not a claim about every historical or contemporary labor-sharing institution. It is a fully enumerated operational mechanism: members contribute labor in rotation, group memory records recognized participation, and elders or recognized stewards act as authorities for recognition, dispute handling, remedy, and re-entry.

The purpose of the witness is to show that the framework does not require a written ledger, token ledger, or blockchain ledger. At the chosen abstraction level, recognized group memory, witness testimony, beneficiary acknowledgment, and steward attestation constitute the relevant state and evidence variables.

Source specification. Let the source state be

$$s = (M, E, Q, C, O, A, D, R, G),$$

where:

- M is the set of recognized members.
- $E \subseteq M$ is the set of elders or recognized stewards.
- Q is the remembered turn order or reciprocal priority relation.
- C records recognized completed labor contributions.
- O records outstanding labor obligations or expected future contributions.
- A records attendance, absence, or partial attendance status.
- D records disputed memory claims.
- R records remedies, make-up obligations, forgiveness, suspension, or re-entry status.
- G records governance permissions, including elder authority and accepted evidence rules.

The state C need not be a written ledger. It may be constituted by group memory, beneficiary acknowledgment, witness recognition, or steward attestation, provided that the relevant contribution, obligation, and remedy effects are observable at the chosen level of abstraction.

Action set. Let the action set be

$$A_{\text{ROLA}} = \{\text{join, schedule, contribute, recognize, allocate,} \\ \text{record-absence, claim-owed, dispute, assign-makeup,} \\ \text{forgive, suspend, re-enter}\}.$$

Routine actions include scheduling a work round, contributing labor, recognizing completed labor, and allocating the current labor benefit to the scheduled beneficiary. Repair actions include disputing remembered participation, assigning make-up labor, forgiving an absence, suspending a member, and re-entering ordinary participation.

Guards. The ordinary guards are:

$$\begin{aligned} g_{\text{member}}(i) : \quad & i \in M, \\ g_{\text{beneficiary}}(j, s) : \quad & j \in M \wedge j \text{ is next under } Q, \\ g_{\text{contribute}}(i, j, \ell, s) : \quad & i \in M \wedge j \in M \wedge \ell \text{ is an accepted labor class,} \\ g_{\text{recognize}}(i, j, \ell, s) : \quad & g_{\text{contribute}}(i, j, \ell, s) \\ & \wedge \text{ accepted evidence confirms contribution.} \end{aligned}$$

Accepted evidence may include beneficiary acknowledgment, sufficient witness memory, steward attestation, or an agreed combination of these. A contribution that lacks accepted evidence does not settle directly; it enters the dispute state D .

Repair guards are:

$$\begin{aligned} g_{\text{absence}}(i, j, s) : \quad & i \in M \wedge j \in M \\ & \wedge i \text{ was expected but not recognized as present,} \\ g_{\text{dispute}}(i, x, s) : \quad & i \in M, \\ & x \text{ concerns a contested contribution, obligation, or turn,} \\ g_{\text{steward}}(e, s) : \quad & e \in E \wedge e \text{ has authority under } G. \end{aligned}$$

Priority rule. Let Π_{ROLA} order claims and remedies as follows:

$$\begin{aligned} & \text{emergency stewarded need} \succ \text{scheduled beneficiary under } Q \\ & \succ \text{recognized make-up obligation} \\ & \succ \text{ordinary future turn.} \end{aligned}$$

If the modeled association does not permit emergency reprioritization, the first layer is omitted. If emergency reprioritization depends on open-ended steward judgment, the routine mechanism remains directly realizable but the reprioritization path is repair-extended.

Settlement operator. The settlement operator Σ_{ROLA} maps expected labor, available contributors, task class, and evidence into one of five observable outcomes:

$$\Sigma_{\text{ROLA}} : (O, C, A, D, R) \rightarrow \{\text{completed, partial, absent,} \\ \text{disputed, remedied}\}.$$

For a recognized completed contribution,

$$\Sigma_{\text{ROLA}}(i, j, \ell) = \text{completed}$$

updates C to record that member i contributed labor class ℓ for beneficiary j , reduces or satisfies the relevant obligation in O , and preserves or advances the turn relation Q .

For an absence,

$$\Sigma_{\text{ROLA}}(i, j, \ell) = \text{absent}$$

updates A and creates a make-up or remedy entry in R .

For a contested memory claim,

$$\Sigma_{\text{ROLA}}(i, x) = \text{disputed}$$

updates D and suspends the contested settlement effect until a steward-authorized repair transition occurs.

Target mapping into (R, V, L, F, H, G) . The memory-based ROLA maps into the pool interfaces as follows:

- R : recognized members, elders or other stewards, accepted labor classes, accepted evidence types, and admissible contribution claims.
- V : task-equivalence rules, such as one workday, half-day, skilled labor, substitute labor, or service-equivalent contribution.
- L : turn-order limits, maximum pull per beneficiary, attendance bounds, emergency caps, and make-up windows.
- F : non-monetary exchange logic, such as hosting duties, food contributions, transport support, or zero-fee operation.

- H : recognized completed labor claims, outstanding labor obligations, make-up obligations, and group-memory attestations.
- G : steward authority, group meeting rules, dispute procedures, forgiveness, suspension, re-entry, and parameter updates.

The holdings state H is not required to be a written ledger. In this witness, H is the operational memory state recognized by the group and made effective through beneficiary acknowledgment, witness recognition, and steward attestation.

Primitive factorization. Routine transitions factorize into the primitive basis U :

`join` $\mapsto$ `issue`

for recognition of membership or participation standing;

`contribute` + `recognize` $\mapsto$ `settle`

for recognized labor fulfillment;

`recognize completed labor` $\mapsto$ `issue`

for a past-oriented certification that a contribution occurred;

`allocate` $\mapsto$ `transfer`; `settle`

for moving the current labor benefit to the scheduled beneficiary;

`record-absence` + `assign-makeup` $\mapsto$ `default/remedy`

for a missed contribution and its remedy.

Repair transitions factorize into U^r :

`dispute` $\mapsto$ `adjudicate`,

`assign-makeup` $\mapsto$ `relimit`,

`forgive` $\mapsto$ `override`,

`re-enter` $\mapsto$ `re-enter`.

Example routine trace. Consider four households A, B, C, D and elder set E . The initial state is:

$$s_0 : M = \{A, B, C, D\}, \quad Q = (A, B, C, D), \quad C = \emptyset, \quad O = \emptyset, \quad D = \emptyset.$$

Round one assigns the benefit to household A :

$$a_1 = \text{schedule}(A).$$

Members B, C, D contribute one accepted workday each to A :

$$\begin{aligned} a_2 &= \text{contribute}(B, A, 1), \\ a_3 &= \text{contribute}(C, A, 1), \\ a_4 &= \text{contribute}(D, A, 1). \end{aligned}$$

The beneficiary and witnesses recognize the contributions:

$$a_5 = \text{recognize}(\{B, C, D\}, A, 1).$$

The state updates to:

$$C' = C \cup \{(B, A, 1), (C, A, 1), (D, A, 1)\},$$

and the turn order advances:

$$Q' = (B, C, D, A).$$

At the macro-step boundary, the externally relevant observables are: A received the labor benefit, B, C, D are recognized as having contributed, no make-up obligation exists, and the next beneficiary is B .

Example repair trace. In round two, household B is the scheduled beneficiary. Household C is expected to contribute but is not recognized as present:

$$a_6 = \text{record-absence}(C, B, 1).$$

The absence guard creates a remedy state:

$$R' = R \cup \{\text{makeup}(C, B, 1, w)\},$$

where w is the make-up window.

If C disputes the absence,

$$a_7 = \text{dispute}(C, \text{absence}),$$

then the claim enters D and ordinary settlement of the contested obligation is paused. A recognized steward applies the repair guard:

$$g_{\text{steward}}(e, s) = 1.$$

The steward-authorized repair transition either confirms absence, recognizes prior contribution, assigns substitute labor, forgives the obligation, or suspends/re-enters the member according to G .

For example:

$$a_8 = \text{assign-makeup}(e, C, B, 1, w)$$

updates R and permits re-entry after completion:

$$a_9 = \text{re-enter}(C).$$

At the macro-step boundary, observable state records: whether C is in good standing, whether a make-up obligation remains, whether B is still owed labor, and whether ordinary rotation resumes.

Classification. The routine labor-rotation mechanism is directly realizable when membership, accepted labor classes, turn order, contribution recognition, and ordinary remedies are explicit. The no-ledger feature does not by itself block realization, because group memory, witness recognition, and steward attestation can serve as state and evidence variables.

The contested-memory and steward-mediated exception paths are repair-extended unless the steward decision rules are fully specified as ordinary guards. If stewards can revise obligations, forgive absences, reorder turns, or redefine acceptable contribution through open-ended judgment, those transitions are not direct. They are repair-extended when the trigger, steward permission, state update, receipt or public recognition, and re-entry or terminal condition are specified.

The mechanism becomes externally anchored only if decisive repair authority leaves the group and is exercised by a court, state official, police authority, or other institution outside the modeled pool-route system.

Thus this witness is classified as:

direct for routine labor rotation;
 repair-extended for contested memory and steward-mediated exceptions.

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Author contributions

William O. Ruddick: conceptualization, methodology, formal analysis, writing, visualization, and revision.

Conflict of interest

The author is affiliated with Grassroots Economics Foundation, whose software repositories and protocol documentation are cited as implementation context. The formal claims, theorem, classifications, and conclusions of the paper do not depend on those artifacts. The author declares no other competing interests.

Data availability statement

This paper does not rely on an external dataset or software package for its formal results. The operational realization test, worked examples, figures, and tables are contained in the manuscript. Public software repositories and protocol documentation associated with Grassroots Economics and Cosmo-Local Credit provide implementation context only; the theorem and classifications do not depend on those artifacts.

AI use disclosure

Language-model assistance was used for drafting and editing. Formal scoping, claims, and responsibility remain with the author.

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